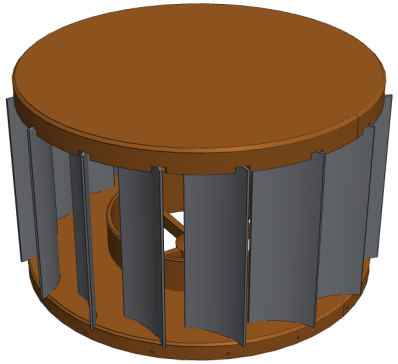
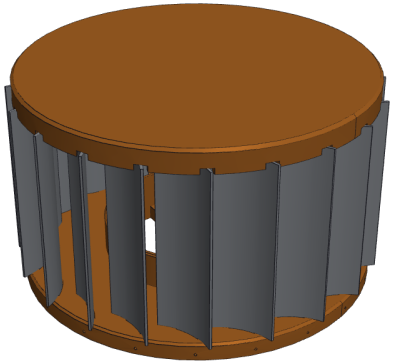
16 / 20 / 24 BLADE COMPARISON - SAME CUPPED BLADE, SAME OD (real CAD renders)

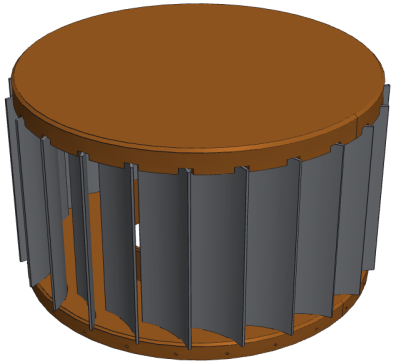
16 blades



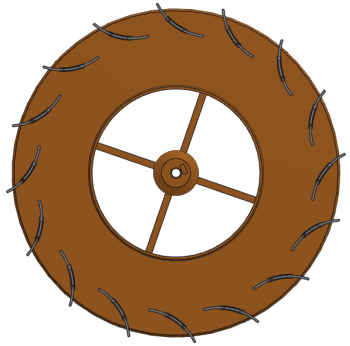
20 blades



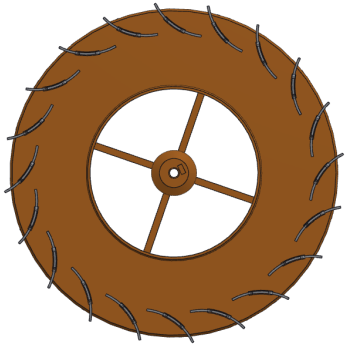
24 blades



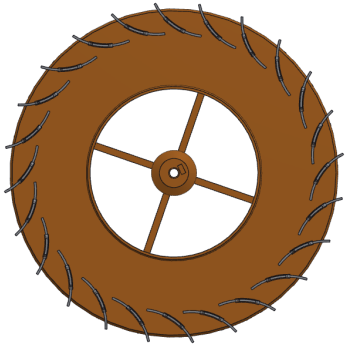
16 blades (upper hub hidden)



20 blades (upper hub hidden)



24 blades (upper hub hidden)



Blades	Chord deg	Cup R	Rotor OD	Air channel	Coverage %	Blade set g	Stock pcs	Targets
16	55	45.0	257.9	18.2	80.3	400.0	4	False
20	55	45.0	257.9	11.5	100.3	500.0	5	True
24	55	45.0	257.9	7.6	120.4	600.0	6	False

Targets: coverage >= 100 % (closed ring, no straight line of sight) AND air channel >= 10 mm.
- 16 blades leave the ring open (~80 %).
- 24 blades choke the air channels (~7.6 mm).
- 20 blades meet both: selected.
- Not a claim of optimal airflow: test (gate F).
- 27-variant table: whirlybird/reports/variant_study.csv

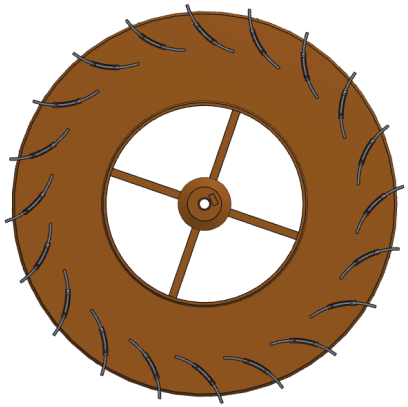
DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
16 / 20 / 24 blade comparison (real CAD)

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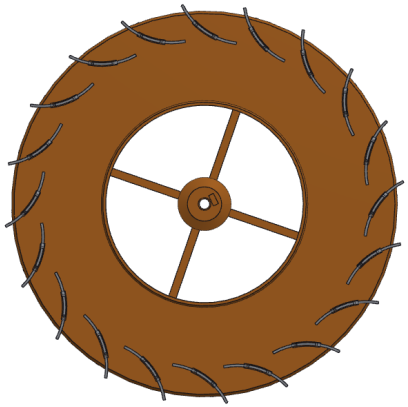
 Source: build123d parametric model + MASTER_PARAMETERS.json

PLAN CHORD ANGLE VARIANTS 45 / 55 / 65 deg (20 cupped blades, upper hub hidden)

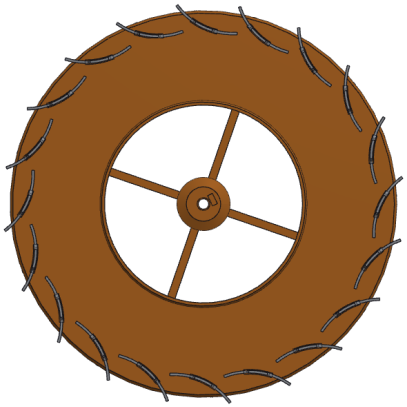
chord angle 45 deg from radial



chord angle 55 deg from radial



chord angle 65 deg from radial



Chord deg	Cup R40: chan / cov%	Cup R45: chan / cov%	Cup R60: chan / cov%
45	14.1 / 89.7	14.6 / 90.8	15.8 / 92.4
55	11.1 / 99.4	11.5 / 100.3	12.6 / 101.9
65	7.7 / 105.8	8.0 / 106.9	8.8 / 108.5

Selected: 55 deg, cup R45: 100 % coverage with an 11.5 mm air channel, 5.4 mm deep scoop.
45 deg leaves line-of-sight gaps; 65 deg narrows the channels below 10 mm. No variant is claimed aerodynamically optimal.

DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Chord angle variants

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

SHEET 2/2 Source: build123d parametric model + MASTER_PARAMETERS.json